

# Wenwen Li

Ph.D. Student, Computational and Applied Mathematics

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## Research Profile

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Statistical and computational methods for inference, learning, and optimization in physical and dynamical systems, combining mathematical guarantees with numerical experiments.

**Focus:** Scientific machine learning; Bayesian inverse problems; data assimilation; uncertainty quantification; Gaussian processes and kernel methods; operator learning.

## Technical Skills

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**Programming:** Python, R, MATLAB.

**Inference & learning:** Bayesian inversion; uncertainty quantification; Gaussian processes; Whittle–Matérn priors; Bayesian optimization; vector-field and solution-map learning.

**Analysis & computation:** PDE and dynamical-system stability; nudging; finite elements; pCN MCMC; convergence and regret analysis; numerical validation.

## Education

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### University of Chicago

Sept. 2024–Present

Ph.D. student, Committee on Computational and Applied Mathematics

Advisor: [Daniel Sanz-Alonso](#) | Expected graduation: **06/2029**

### Sichuan University

Sept. 2019–June 2024

B.S., Mathematics and Applied Mathematics

Class rank: **1/104** | Average score: **92.3/100**

## Research Experience

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### Doctoral Research, University of Chicago

2024–Present

#### Data assimilation with learned surrogate dynamics

With [Daniel Sanz-Alonso](#); submitted to *Physica D*, 2026

- Developed a finite-dimensional analysis of continuous data assimilation using learned dynamics to estimate system states from partial observations.
- Established exponential tracking up to an explicit error floor that separates surrogate approximation error from observation noise, quantifying when learned models support reliable state estimation.
- Connected vector-field and short-time solution-map learning to tracking guarantees and training-data requirements; validated the theory on Lorenz–96. [Python reference code](#).

#### Bayesian inverse problems on metric graphs

With [David Bolin](#) and [Daniel Sanz-Alonso](#); *Inverse Problems*, 2026

- Formulated Bayesian recovery of diffusion coefficients in elliptic and fractional elliptic equations on networks from noisy solution measurements.
- Established forward-model stability and Bayesian well-posedness using metric-graph regularity theory and Gaussian Whittle–Matérn priors.
- Evaluated coefficient reconstruction and uncertainty quantification using finite-element PDE solvers and pCN MCMC. [R reference code](#).

## Research Experience (continued)

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### Bayesian optimization on networks

With [Daniel Sanz-Alonso](#) and [Ruiyi Yang](#); *Journal of Computational Physics*, 2026

- Developed sequential Bayesian optimization for expensive black-box objectives on metric graphs, using Gaussian process surrogates adapted to network geometry.
- Established regret bounds and analyzed unknown objective smoothness and finite-element representations of the Whittle–Matérn prior.
- Evaluated the methods on synthetic graphs and maximum a posteriori estimation for Bayesian inversion on a telecommunication network.

### Research Intern, University of British Columbia

July–Nov. 2023

Supervisor: [Christoph Ortner](#) | Mitacs Globalink

- Analyzed stability of inverse problems in the linear atomic cluster expansion (ACE), using parameterized interatomic-potential models.
- Studied finite-dimensional well-posedness, Gram-matrix expansions, and covariance and higher-order error terms through basis expansions and asymptotic analysis.

## Publications & Manuscripts

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### Published journal articles

1. **Wenwen Li**, [Daniel Sanz-Alonso](#), and [Ruiyi Yang](#). Bayesian Optimization on Networks. *Journal of Computational Physics*, 558:114819, 2026. [arXiv:2510.27643](#).
2. [David Bolin](#), **Wenwen Li**, and [Daniel Sanz-Alonso](#). Elliptic Bayesian Inverse Problems on Metric Graphs. *Inverse Problems*, 2026. [arXiv:2507.18951](#).

### Submitted manuscript

3. **Wenwen Li** and [Daniel Sanz-Alonso](#). Continuous Data Assimilation with Learned Surrogate Dynamics. Submitted to *Physica D*; preprint [arXiv:2606.00480](#), 2026.

### Work in preparation

*Adaptive Bayesian Multiscale Learning of Linear Operators*. Manuscript in preparation.

## Selected Research Presentations

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### Kernel Approximation and Gaussian Processes, BIRS (Banff)

Sept. 2026

Bayesian optimization on networks.

### SIAM Mathematics of Planet Earth (MPE26)

July 2026

Continuous data assimilation with machine-learned surrogates.

### SIAM Annual Meeting (AN26), Cleveland

July 2026

Bayesian Optimization on Networks with Application in Inverse Problems.

### AMS Western Sectional Meeting, Boise

March 2026

Elliptic Bayesian inverse problems on metric graphs.

### SIAM Analysis of Partial Differential Equations (PD25)

Nov. 2025

Bayesian inverse problems on metric graphs.

## Honors & Awards

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Mitacs Globalink Research Internship Award

2023

Premium Scholarship, Sichuan University

2023

China National Scholarship for Undergraduates

2021